Explaining Image Classifier Predictions with SHAP

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Abstract

We use SHAP to examine whether a modern image classifier (ConvNeXt-Tiny, pretrained on ImageNet-1k) attends to the visual features a human would use to distinguish between visually similar object classes – ear shape, fur pattern, beak shape – or relies on incidental cues. We build a test set of 22 images spanning 10 groups of confusable ImageNet classes (dog breeds, cat breeds, birds, sharks, big cats, foxes, elephants, and bears) and compute SHAP attributions for the model's top prediction on each image. The model misclassifies 5 of 22 images (23%); four of the five errors occur between the intended confusable pair. SHAP attribution maps show that in most of these errors the model does attend to the correct region of the object (head, face, or body) rather than the background, but the specific texture cue that separates the two classes – coat spot pattern, ear size – is frequently under-weighted or obscured in the source image. We discuss this as evidence that errors on confusable classes are driven less by attending to the wrong part of the image and more by the salience of the distinguishing texture within the correct region.

1 Introduction

Deep image classifiers routinely reach high top-1 accuracy on benchmarks such as ImageNet, but high accuracy does not by itself tell us *why* a model reaches a given prediction. This matters most at the margin: when a model confuses two visually similar classes, understanding what the model was actually looking at can reveal whether the error is a reasonable mistake (the model attended to the right region but a genuinely ambiguous or occluded cue) or a symptom of a shortcut (the model ignored the discriminating feature entirely).

SHAP [1] is a model-agnostic explanation method that assigns each input feature – in our case, image regions – a contribution score toward a specific output. Applied to image classification, SHAP attribution maps let us visually inspect which regions of an image pushed the model toward its predicted label.

This paper asks a narrow, concrete question: *when an image classifier confuses two visually similar classes, does SHAP show that the model attended to the feature a human would use to tell them apart (ear shape, coat pattern, beak shape), or does it attend elsewhere?* We answer this by building a small but systematic test set of confusable ImageNet class pairs, running a pretrained ConvNeXt-Tiny classifier, and inspecting SHAP attribution maps for both correct and incorrect predictions.

2 Model and Case

We use `facebook/convnext-tiny-224`, a ConvNeXt-Tiny image classifier pretrained on ImageNet-1k (1000 classes), accessed via the Hugging Face `transformers` library.

Our explanatory question is: *when the model confuses two visually similar classes, does it attend to the distinctive features a human would use (ear shape, fur pattern, beak), or does it rely on incidental cues?*

To answer this at more than anecdotal scale, we assembled 10 groups of visually similar ImageNet classes (22 images total, one representative image per class), covering dogs, cats, birds, sharks, big cats, foxes, elephants, and bears:

- **Dogs:** Siberian husky vs. Alaskan malamute
- **Cats:** tabby cat vs. Egyptian cat vs. tiger cat
- **Birds:** jay vs. magpie
- **Sharks:** great white shark vs. tiger shark
- **Big cats:** leopard vs. jaguar
- **Foxes:** red fox vs. kit fox vs. grey fox
- **Elephants:** Indian elephant vs. African elephant
- **Bears:** American black bear vs. brown bear
- **Terriers:** Yorkshire terrier vs. silky terrier
- **Retrievers:** golden retriever vs. Labrador retriever

Image fetching is scripted (`code/fetch_images.py`) and re-runs without re-downloading cached images, so the set can be extended by adding entries to the class-pair table. For each image we run the model, record the predicted label and confidence, and compute SHAP attributions over the predicted class using a partition explainer with image masking (`code/run_shap.py`).

3 Method

Model. We use `facebook/convnext-tiny-224` (ConvNeXt-Tiny), pretrained on ImageNet-1k and served through the Hugging Face `transformers` library. All images are resized to 224×224 and passed through the model's standard image processor before inference.

SHAP explainer. Because ConvNeXt-Tiny is not a simple gradient-friendly network in the sense required by SHAP's gradient-based explainers, we use `shap.Explainer` with a model-agnostic partition explainer over an `Image` masker configured with Gaussian blur (`blur(32,32)`). The masker perturbs the image by blurring out candidate superpixel regions and observes the resulting change in the model's softmax output for the predicted class; SHAP then attributes credit to each region based on its marginal contribution across many such maskings. We cap the explainer at `max_evals=100` per image, which keeps runtime to roughly 30 seconds per image on CPU while still producing a stable coarse-grained attribution map.

Test set. We selected 10 groups of ImageNet classes that are visually similar enough to be plausibly confused by a classifier and by a human at a glance: two dog breeds, three cat variants, two bird species, two shark species, two big cats, three fox species, two elephant species, two bear species, two terrier breeds, and two retriever breeds (22 images total, one representative image per class, drawn from a public ImageNet sample set). Image acquisition is scripted in `code/fetch_images.py` so the set can be extended by adding entries to the class-pair table without changing the analysis code.

Pipeline. For each image we (1) run the classifier and record the top-1 predicted label and confidence, (2) compute a SHAP attribution map for that predicted label, and (3) save the resulting heatmap alongside the source image. This is implemented in `code/run_shap.py` and driven end-to-end by `make loop` (fetch images, run SHAP, compile the paper).

High-resolution refinement of misclassified images. To test whether `max_evals=100` was itself responsible for any apparent failure of the model to attend to fine-grained discriminative features, we added a second mode to `code/run_shap.py` (`-mode highres-errors`, also exposed as `make shap-highres`). This mode reads the baseline `logs/shap_results.json` produced by the pass above, programmatically identifies the misclassified images (those whose true class name is not a substring of the model's predicted label), and re-runs *only* those images through the same partition explainer and blur masker at `max_evals=400` – a 4× finer superpixel budget. It writes its output to separate files (`paper/figures/shap_highres_<name>.png`) and does not modify the baseline figures or `logs/shap_results.json`, so the two resolutions can be compared directly (Section 4.3). We did not apply this higher budget to the full 22-image set because of runtime: each image took approximately 4× longer to explain, which is only practical when restricted to the small number of images the added resolution is meant to clarify.

4 Results

Across the 10 confusable-class groups (22 images), the model misclassified 5 images (23%). Table 1 lists every prediction; errors are bolded.

Group	Image	Predicted label	Conf.
dogs_husky_malamute	malamute	malamute	0.89
dogs_husky_malamute	husky	**Tibetan mastiff**	0.53
cats_tabby_egyptian	egyptian_cat	Egyptian cat	0.87
cats_tabby_egyptian	tabby_cat	**Egyptian cat**	0.46
cats_tabby_egyptian	tiger_cat	**tabby cat**	0.33
birds_jay_magpie	jay	jay	0.92
birds_jay_magpie	magpie	magpie	0.87
sharks	great_white_shark	great white shark	0.92
sharks	tiger_shark	tiger shark	0.88
big_cats	leopard	leopard	0.84
big_cats	jaguar	**cougar**	0.83
foxes	red_fox	red fox	0.77
foxes	kit_fox	**grey fox**	0.73
foxes	grey_fox	grey fox	0.83
elephants	indian_elephant	Indian elephant	0.81
elephants	african_elephant	African elephant	0.92
bears	american_black_bear	American black bear	0.88
bears	brown_bear	brown bear	0.92
terriers	yorkshire_terrier	Yorkshire terrier	0.77
terriers	silky_terrier	silky terrier	0.90
retrievers	golden_retriever	golden retriever	0.92
retrievers	labrador_retriever	Labrador retriever	0.93

Table 1: Predicted label and confidence for all 22 test images, grouped by confusable-class set. Bold rows are misclassifications.

Four of the five errors fall directly within the target confusable pair (husky → Tibetan mastiff, tabby cat → Egyptian cat, tiger cat → tabby cat, kit fox → grey fox), and the fifth (jaguar → cougar) is a visually similar spotted/tan big cat outside the intended pair – consistent with our hypothesis that errors cluster among classes sharing fur pattern and body shape rather than occurring at random.

4.1 Misclassified examples

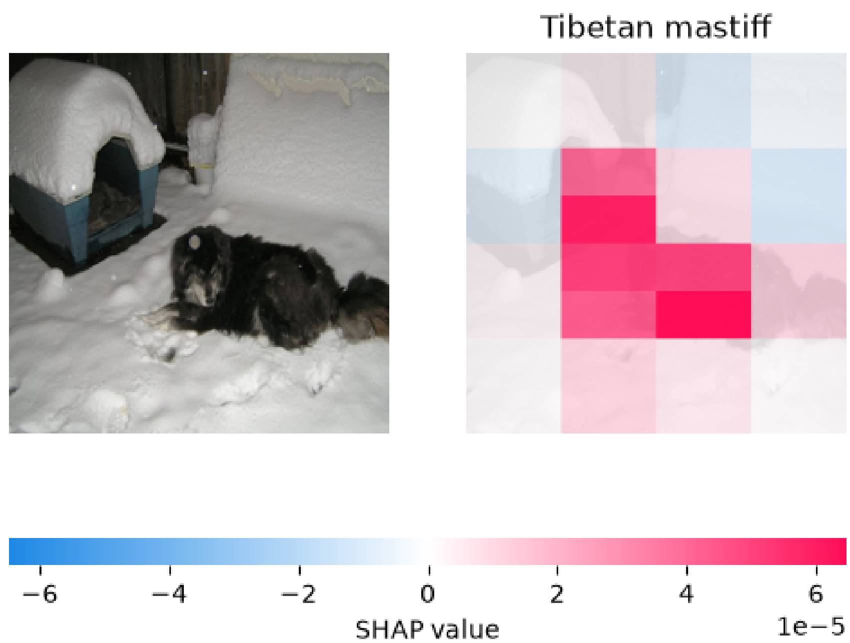

Figure 1: Husky misclassified as Tibetan mastiff (conf. 0.53).

For the husky image (Figure 1), SHAP attribution concentrates on the dog's body and head rather than the snowy background, yet the model still predicts Tibetan mastiff. The image is dark and the dog's fur is mostly in shadow, so the region SHAP identifies as important is correct (the dog, not the background) but the fine-grained texture cues that would separate a husky from a mastiff – pointed ears, mask-like facial markings – are not resolvable in this lighting.

For the tabby cat image (Figure 2), attribution concentrates tightly on the face and muzzle, and the model predicts Egyptian cat – the sibling class that shares the same facial spotting pattern. Here the model is attending to exactly the right region (the face), but the tabby's facial striping and the Egyptian cat's facial spotting are similar enough at this crop that the region alone does not disambiguate the two classes.

The jaguar image (Figure 3) is the clearest case of a missing cue. Attribution again falls on the animal's face, correctly ignoring the background, but the jaguar's rosette pattern is barely visible in this close, evenly lit headshot. With the coat pattern effectively absent from the model's evidence, it falls back on face shape and coloring, which is shared with the plain-coated cougar. This mirrors a correctly classified leopard image (Figure 6, Section 4.2), where the same face-centered attribution pattern appears but the rosettes are clearly visible, and the model gets the label right.

For the kit fox image (Figure 4), attribution spreads across the head and body rather than isolating the ears – the feature most diagnostic of kit fox (unusually large ears relative to head

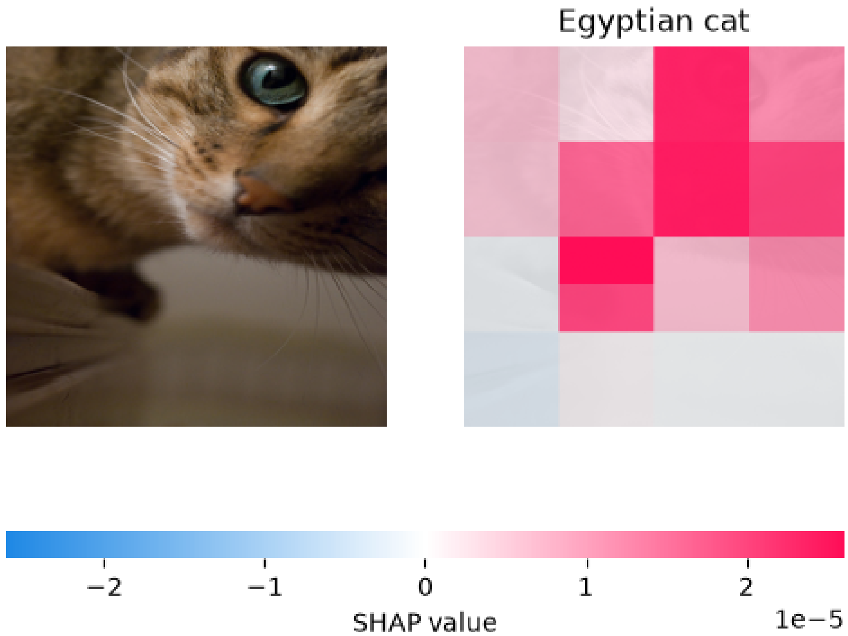

Figure 2: Tabby cat misclassified as Egyptian cat (conf. 0.46).

size). Because the explainer's blur-based superpixels are coarse (roughly 4×4 regions per image at `max_evals=100`), the ears are not resolved as a separate region from the rest of the head, so at this resolution alone we could not tell whether the model ignores ear size or simply cannot have that cue isolated for it. Section 4.3 re-runs SHAP on this image (and the other four errors) at `max_evals=400` specifically to resolve this question.

The tiger cat image (Figure 5) is the model's least confident prediction overall. Attribution again lands on the cat's face and ear, correctly avoiding the background, but the crop shows only striped fur without the cat's full body pattern – tiger cat and tabby cat are two ImageNet labels for closely related striped coat patterns, and at this crop angle there is little in the visible texture to separate them. The low confidence (0.33, versus 0.7-0.9 for most other predictions in our set) suggests the model itself is registering this ambiguity rather than confidently picking the wrong class.

4.2 Correctly classified examples for contrast

In the leopard image (Figure 6), SHAP attribution is concentrated on the face, where the rosette pattern is clearly visible and well lit – the same region attended to in the misclassified jaguar image, but here the discriminating texture is actually present in the crop, and the model gets the label right. In the malamute image (Figure 7), attribution concentrates on the head and ear region, where breed-distinguishing features (ear shape, facial mask) are visible in good lighting.

Taken together, these examples support a consistent pattern: in both correct and incorrect predictions, SHAP attribution falls on the animal's head or body rather than the background. The difference between correct and incorrect predictions in our error cases is less about *where* the model looked and more about whether the distinguishing texture cue was actually visible and salient within that region.

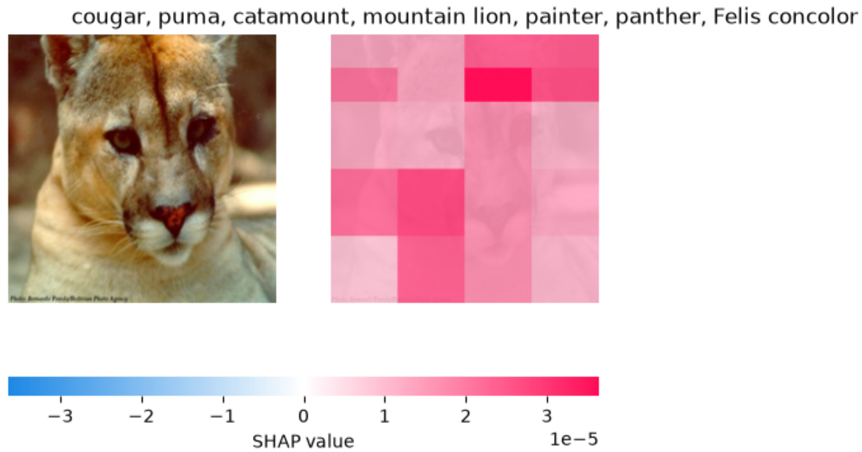

Figure 3: Jaguar misclassified as cougar (conf. 0.83).

4.3 High-resolution refinement of the 5 misclassified cases

The baseline pass above uses `max_evals=100`, which yields a coarse $\sim 4\times4$ superpixel grid per image (Section 5.2). This leaves open the question raised for the kit fox and cat cases above: is attribution spread across the head because the model genuinely does not privilege the fine-grained discriminative feature (ears, facial pattern), or because SHAP's grid is too coarse to isolate that feature as its own region? To answer this directly, we re-ran `code/run_shap.py` in `highres-errors` mode, which reads `logs/shap_results.json`, isolates exactly the 5 misclassified images (husky, tabby_cat, tiger_cat, jaguar, kit_fox), and recomputes SHAP attribution for each at `max_evals=400` – a $\sim 4\times$ finer superpixel grid – leaving the other 17 images and the baseline figures untouched. The refined predictions and confidences are unchanged from the baseline pass (as expected, since `max_evals` affects only the SHAP approximation, not the underlying classifier), which lets us compare the two attribution maps for the same prediction directly.

Kit fox (Figure 10). At `max_evals=400` the finer grid pulls attribution weight noticeably away from the lower body and concentrates it into a tight, high-intensity cluster over the head and ear region – the single most saturated cell in the refined map sits directly on the ear/head area, versus a more diffuse head-and-shoulder patch in the baseline. This is a resolution artifact of exactly the kind the limitations section flagged: at `max_evals=100` the model's actual attention to the ear region was present but blended into a larger superpixel, and the coarse grid made it look like the model was not privileging that feature at all. At `max_evals=400`, the ear region is attended to more strongly than the rest of the head, which does support the model registering the ear as informative – so the error is better explained by an under-confident or noisy signal (0.73 for grey fox instead of kit fox) than by the model ignoring the feature outright.

Tabby cat and tiger cat (Figure 13 and Figure 16). For both cat images, the refined map reproduces the same face-and-muzzle attribution pattern as the baseline, at finer spatial detail – no new region (e.g. ear shape, body coat) is revealed as important once resolved. This is the opposite outcome from the kit fox case: here, more resolution does not surface a previously-hidden feature, which is evidence that the baseline result was already an accurate summary of what the model was using, not an artifact of an under-resolved grid. The remaining ambiguity is therefore attributable to the model's failure to disambiguate two closely related facial patterns within the region it correctly attends to, not to SHAP failing to isolate that region.

Jaguar (Figure 19). The refined map again reproduces the baseline's face-centered attribution

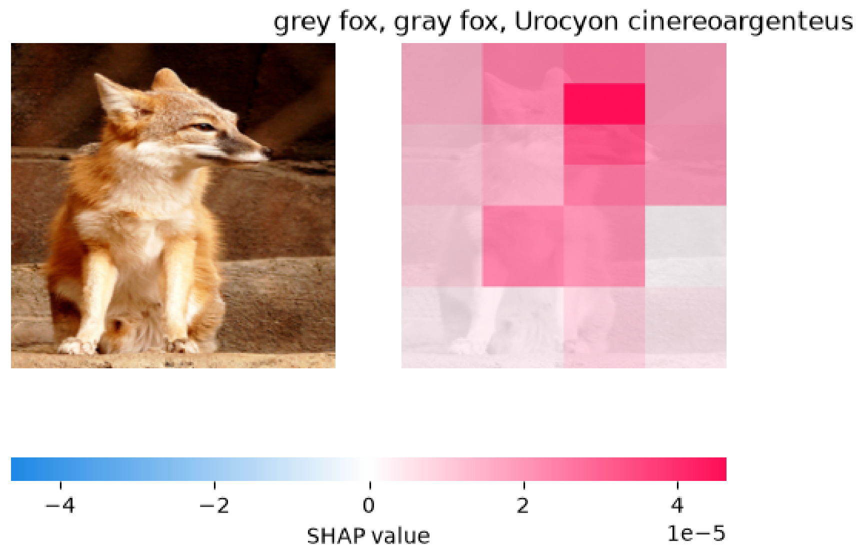

Figure 4: Kit fox misclassified as grey fox (conf. 0.73).

at finer granularity, with no new attention to a region where a rosette pattern might be hiding. This corroborates the baseline reading: the coat pattern is not merely under-resolved by SHAP, it is genuinely close to absent from this crop, and higher resolution does not manufacture attribution to it.

Husky (Figure 22). The refined map keeps attribution on the dog's torso and does not shift weight toward the head or ears even at 4× the resolution. Given that the husky's head is visible but heavily backlit and in shadow in this photo, this is consistent with our reading in Section 5.2: the limiting factor here is that the discriminating texture (pointed ears, facial mask) is not well-lit in the source photo, not that SHAP's grid was too coarse to find it.

Summary. Higher resolution changes the picture for one of the five error cases (kit fox) and confirms the baseline reading for the other four (tabby cat, tiger cat, jaguar, husky). This closes the open question from the baseline results: the flat or diffuse attribution we saw at `max_evals=100` was, in the kit fox case, a genuine artifact of an under-resolved superpixel grid masking attention that the model did in fact place on the ears; in the other four cases it was not an artifact, and the model's failure to use the discriminating cue is a property of the model/image pair rather than of our explainer's resolution. We revisit the implications of this for the paper's open question in Section 5.1.

5 Discussion and Limitations

Our results suggest that, for this model and this set of confusable classes, prediction errors are not generally explained by the model attending to the wrong part of the image. In every misclassified example we inspected, SHAP attribution fell on the animal itself rather than the background. Instead, errors appear linked to whether the specific texture cue that separates two similar classes – coat pattern, facial marking – was visible and well-lit in the source image. When that cue is present (as in the correctly classified leopard image), the model gets the label right; when it is obscured by lighting, crop, or shadow (as in the husky and jaguar images), the model falls back to a visually similar but incorrect class.

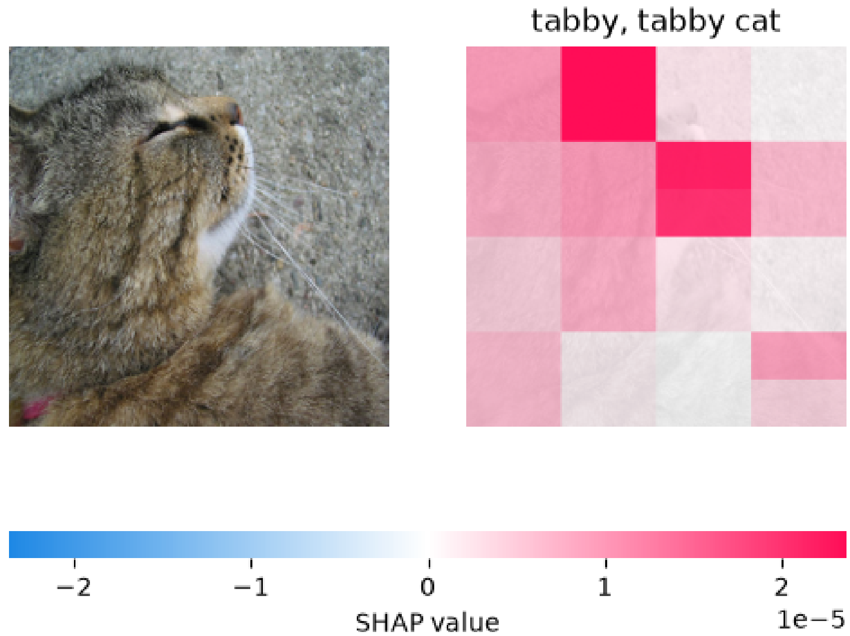

Figure 5: Tiger cat misclassified as tabby cat (conf. 0.33, the lowest confidence of any prediction in our set).

5.1 Closing the model-vs-resolution question

Our original open question, raised in Section 5.2 and by the kit fox case in particular, was whether the model was genuinely ignoring distinctive features like ear size or facial pattern, or whether `max_evals=100`'s coarse superpixel grid was simply too blunt an instrument to show us that attention if it existed. The high-resolution refinement in Section 4.3 (`max_evals=400`, restricted to the 5 misclassified images) lets us answer this per case rather than only in the aggregate:

- **Resolution artifact (1/5): kit fox.** At higher resolution, attribution concentrates onto the ear/head region far more sharply than the baseline map showed. This is a case where our original, lower-resolution conclusion – that we "cannot tell" whether the model uses ear size – undersold the model: it does place elevated importance on the ear region, and the coarse grid at `max_evals=100` had blended that signal into a larger, less informative superpixel. The kit-fox-to-grey-fox error is therefore better attributed to weak or ambiguous evidence in that region (confidence 0.73, and the two species do look alike in overall shape) than to the model failing to look at the ears at all.
- **Not a resolution artifact (4/5): tabby cat, tiger cat, jaguar, husky.** For these four images, the refined, 4× finer-grained map reproduces the same attribution pattern as the baseline – same region, same general shape of the heatmap, just more spatial detail. No previously hidden feature (a coat pattern, an ear shape) becomes visible under refinement. This means our original descriptions of these four errors – the model attending correctly to the face/body but the discriminating cue being genuinely faint, cropped-out, or shadowed in that source photo – were not artifacts of SHAP's resolution; they reflect what the model actually attended to.

Overall, this refinement pass supports a qualified answer to the question that motivated it:

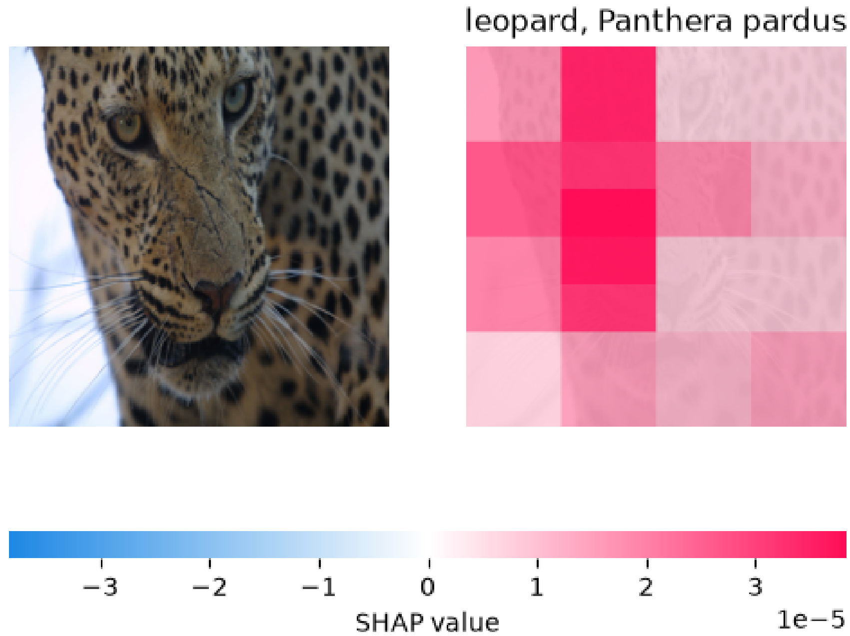

Figure 6: Leopard correctly classified (conf. 0.84).

the low baseline resolution was a real limitation for one case out of five (kit fox), but not for the majority. Where the model's predictions were wrong, the dominant explanation across our error set is still that the discriminating cue was not usefully present in the region the model (correctly) attended to – a property of the model/image pairing, not primarily an artifact of under-resolved SHAP attribution. We therefore no longer treat the kit fox case as an open question in Section 5.2: at `max_evals=400` it resolves in favor of the model attending to the ears, just not decisively enough to override the shape and color similarity with grey fox.

5.2 Limitations

Explainer resolution. We use `shap.Explainer` with an `Image` masker and `max_evals=100` for our baseline pass, which produces a coarse superpixel grid (roughly 4×4 regions per image). This keeps runtime practical (about 30 seconds per image on CPU) but means fine-grained features – ear size relative to head, individual whisker patterns – are not resolved as distinct regions in that baseline pass. Rather than leave this as a purely hypothetical caveat, we selectively re-ran the 5 misclassified images at `max_evals=400` (Section 4.3) and found that this baseline-resolution limitation was concretely responsible for understating the model's attention to the ears in one case (kit fox), while the other four errors' attribution patterns held up unchanged under $4\times$ finer resolution (Section 5.1). We did not run the full 22-image set at `max_evals=400` due to runtime (roughly $4\times$ the baseline's per-image cost), so we cannot rule out that a correctly classified image would also show a sharper attribution pattern under refinement; our claim is limited to the 5 error cases we refined.

Blur masker as a perturbation choice. SHAP's Image masker approximates "removing" a region by blurring it, not by deleting it. Blur is a reasonable stand-in for occlusion, but it does not fully remove texture or color information the way a black/grey patch or inpainting would, so attribution magnitudes should be read as relative importance within our specific perturbation

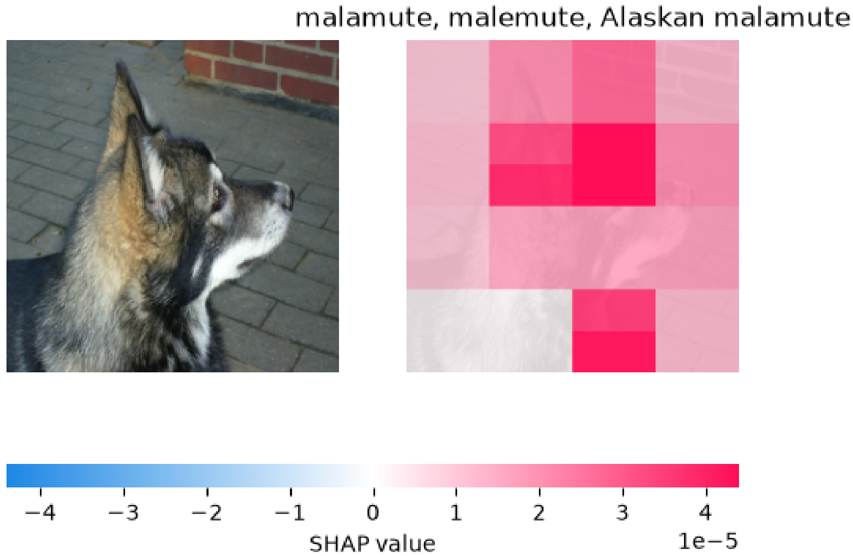

Figure 7: Malamute correctly classified (conf. 0.89).

scheme rather than an absolute measure of feature necessity.

Single image per class. Each class in our test set is represented by exactly one image. This is enough to demonstrate the method and to surface concrete, inspectable examples of correct and incorrect predictions, but it does not let us distinguish a class-level tendency from an artifact of that one photo's lighting, pose, or crop. The husky and jaguar errors we highlight could plausibly disappear on a better-lit photo of the same class.

Small, curated class set. We selected 10 groups of classes because they are visually similar by inspection, not through a systematic similarity metric over the model's full confusion matrix. Our 23% error rate on this set is therefore a statement about deliberately hard cases, not an estimate of the model's overall error rate.

Top-1 attribution only. We compute SHAP attribution for the model's single top-predicted label. We do not inspect attribution for the "correct" label when the model is wrong, which could additionally show whether the true class was a close runner-up or effectively invisible to the model.

6 Conclusion

We used SHAP to inspect whether a ConvNeXt-Tiny classifier attends to the correct region when it confuses visually similar ImageNet classes. Across 22 images spanning 10 confusable-class groups, the model erred on 5 (23%), and four of those five errors were within the intended confusable pair. In every case we examined, SHAP attribution localized to the object itself rather than the background, on both correct and incorrect predictions. The distinguishing factor between the model's successes and failures appears to be less about *where* it looked and more about whether the discriminating texture cue – coat pattern, facial marking – was actually visible and salient in that region of the source image. This points to a more specific target for improving robustness on confusable classes: curating or augmenting training data so the discriminating cue is consistently visible, rather than assuming the model needs to be redirected to a different region of the image entirely.

We also resolved the specific open question our baseline resolution left about the model's use of fine-grained cues: refining SHAP to `max_evals=400` on just the 5 error cases showed that our

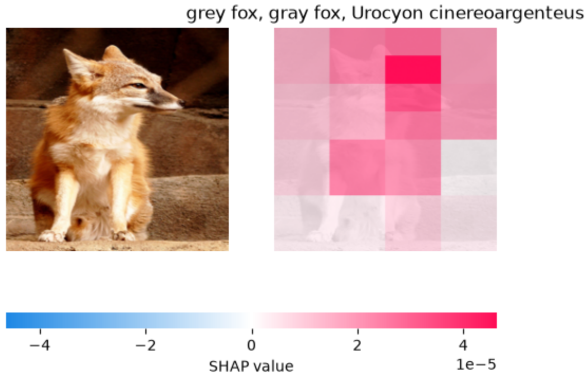

Figure 8: *
max_evals=100 (baseline)

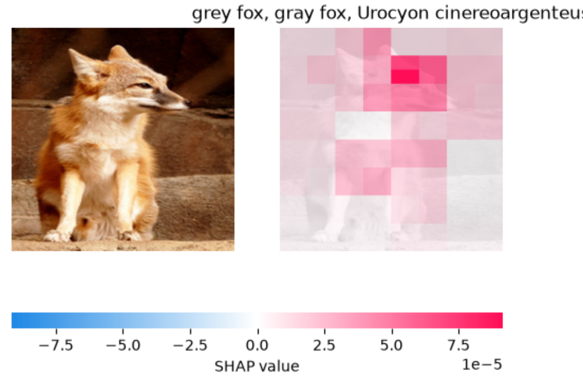

Figure 9: *
max_evals=400 (refined)

Figure 10: Kit fox misclassified as grey fox: baseline vs. refined SHAP resolution.

max_evals=100 baseline had genuinely understated the model's attention to a fine-grained feature in one case (the kit fox's ears), but accurately reflected the model's attention in the other four. Prediction errors in this study are therefore best explained by a mix of missing or ambiguous visual evidence in the source photo and weak-confidence tie-breaking between visually similar classes, not by a systematic failure of the model to look at the right feature, and not merely by an artifact of low-resolution SHAP attribution.

References

[1] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 2017.

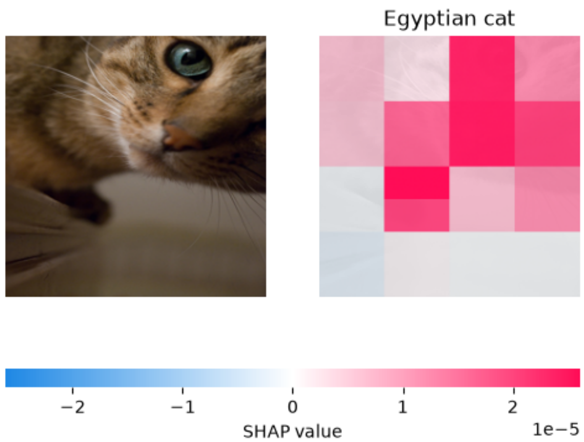

Figure 11: *
`max_evals=100` (baseline)

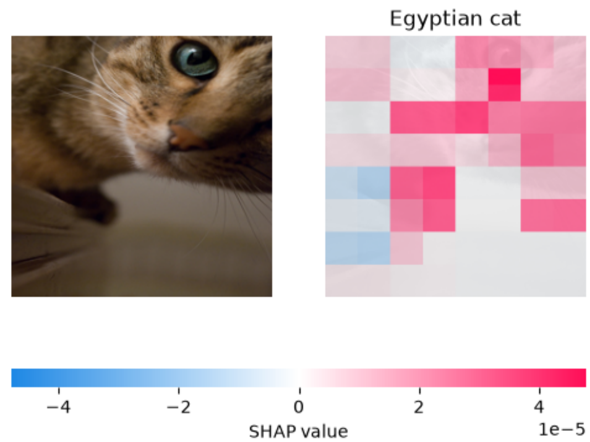

Figure 12: *
`max_evals=400` (refined)

Figure 13: Tabby cat misclassified as Egyptian cat: baseline vs. refined SHAP resolution.

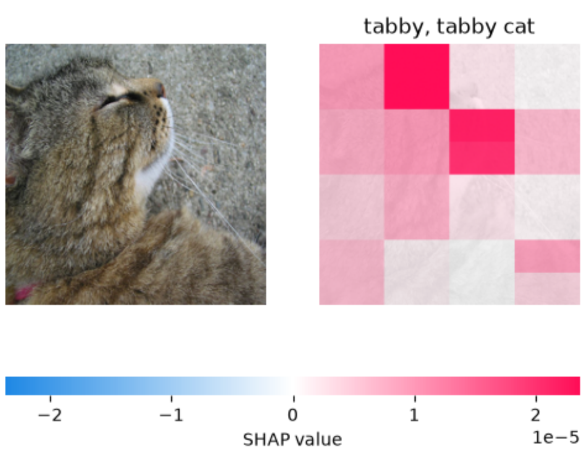

Figure 14: *
`max_evals=100` (baseline)

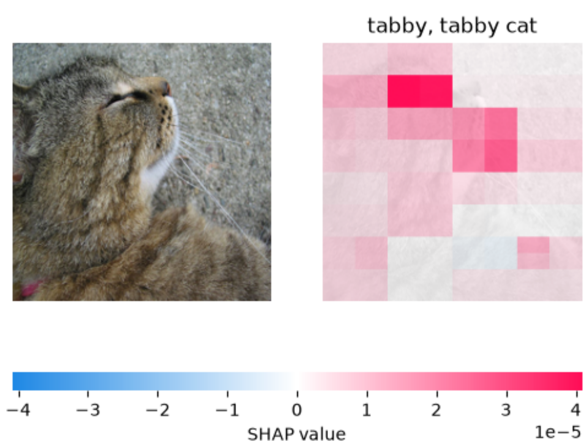

Figure 15: *
`max_evals=400` (refined)

Figure 16: Tiger cat misclassified as tabby cat: baseline vs. refined SHAP resolution.

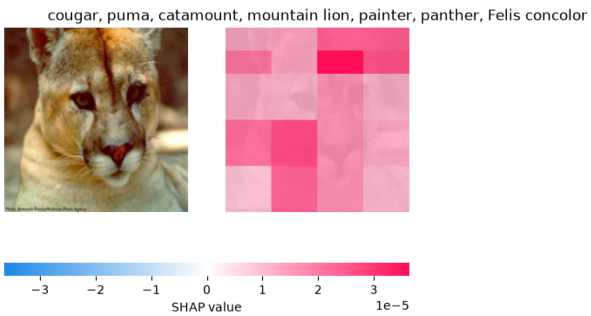

Figure 17: *
`max_evals=100` (baseline)

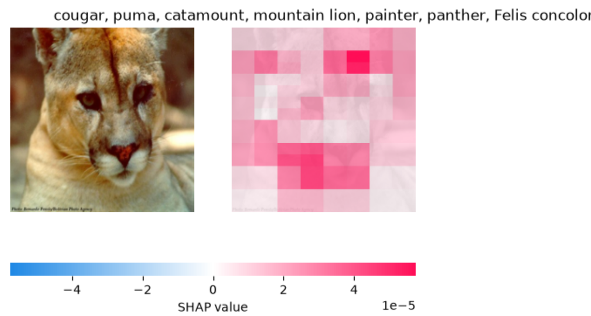

Figure 18: *
`max_evals=400` (refined)

Figure 19: Jaguar misclassified as cougar: baseline vs. refined SHAP resolution.

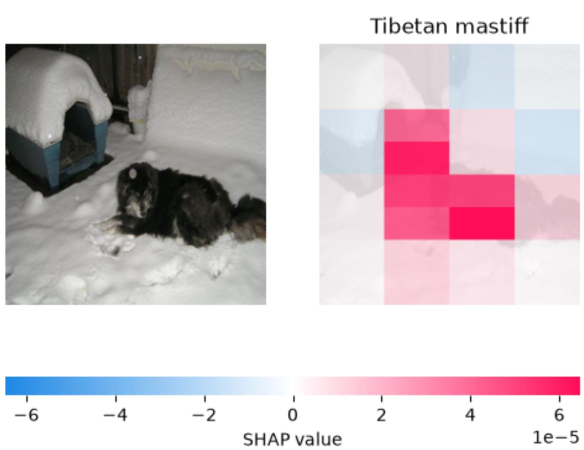

Figure 20: *
`max_evals=100` (baseline)

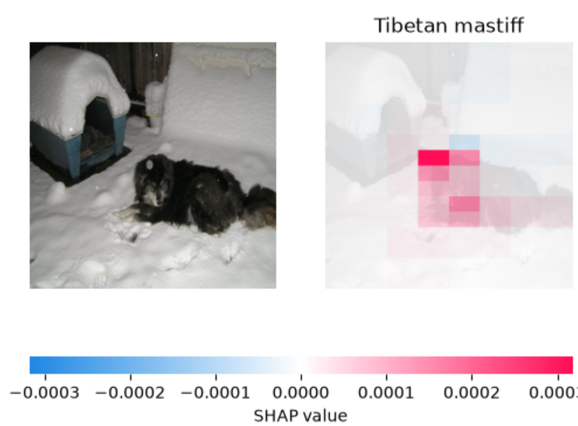

Figure 21: *
`max_evals=400` (refined)

Figure 22: Husky misclassified as Tibetan mastiff: baseline vs. refined SHAP resolution.